

A regional soil classification framework to improve soil health diagnosis and management

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Abstract

Soil health research has developed soil tests to validate management strategies for creating healthier soils, such as cover crops, compost, and reduced tillage. Missing in this research agenda are soil health management strategies and diagnostics tailored to the systematic variation of soils. The Soil Survey Geographic (SSURGO) database offers comprehensive estimates for many properties linked to soil health indicators. This research uses unsupervised classification of soil surface (0–30 cm) properties and root restrictive horizons in an internationally important agricultural region, the Central and Coastal Valleys of California, covering 5.6 million ha of mostly cropland. *K*-means clustering of 10 soil properties derived from SSURGO revealed groups distinct from USDA-NRCS Soil Taxonomy. Clustering metrics, data visualization, and validation tests justified a seven-region soil health conceptual model that divided the landscape into two broad classes—those with and without performance limitations. Results revealed three pedogenic conditions limiting soil performance: (a) root restrictive horizons that impede root growth into deeper soil; (b) salinity and alkalinity harmful to crops; and (c) shrink–swell properties making soils challenging to cultivate. Soils without performance limitations, soils with restrictive horizons, and salt-affected soils are further divided by texture and/or organic matter for a total of seven soil health regions. In reality, variation in soil properties is often continuous across the landscape and sometimes fractal in nature, but this research establishes a testable framework for evaluating soil health diagnostics according to geographically coherent differences in soil properties, demonstrating potential for how soil health management and interpretation can be tailor-made to the landscape.

1 | INTRODUCTION

Soil health has been defined as essentially the soil's capacity to function as a living ecosystem to support plant, ani-

mal, and human life (Doran & Zeiss, 2000). A healthy soil provides numerous services through its multifunctionality, such as processing carbon and storing it as organic matter (OM) through a complex soil food web, cycling nutrients and making them bioavailable to crop uptake, regulating macroporosity and pore connectivity important to infiltration and percolation of water, and controlling species within the soil biotic community, including crop pests (Kibblewhite, Ritz, & Swift, 2008). Soil health is thought to be malleable and

Abbreviations: AWC, available water capacity; BD, bulk density; CEC, cation exchange capacity; EC, electrical conductivity; K_s , saturated hydraulic conductivity; KSSL, Kellogg Soil Survey Laboratory; LEP, linear extensibility percentage; OM, organic matter; SOC, soil organic carbon; SSURGO, Soil Survey Geographic

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manageable by the farmer (Doran, Sarrantonio, & Liebig, 1996), with healthy soils viewed as being more resistant and resilient in response to disturbances, providing ecosystem stability (Griffiths, Bonkowski, Roy, & Ritz, 2001). To date, soil health research has sought to develop analyses of soil indicators that can, for example, shed light on the effect land management practices have on soil health with a focus on validating practices and ensuring long-term soil sustainability (Nunes, Karlen, Veum, Moorman, & Cambardella, 2020). However, this effort, especially interpretation of analyses, has proved challenging.

Given the complexity of soils, it can be difficult to separate the effect of land management practices from the effect of inherent differences in soil forming factors. A North Carolina study of soil health metrics across multiyear agroecosystem trials revealed more apparent soil health differences across physiographic regions (mountain, piedmont, and coastal) than across contrasting management systems (Roper, Osmond, Heitman, Waggoner, & Reberg-Horton, 2017), although a reanalysis of the dataset showed an effect of management on both crop yield and some soil health indicators (van Es & Karlen, 2019). This interpretation challenge is mirrored in the decades-long discussion concerning definitions of soil quality vs. soil health. Though nearly synonymous, soil quality reflects the capability of a soil to perform various functions that arise from both inherent, relatively unchangeable properties (such as texture and mineralogy) and semidynamic properties (such as soil structure and OM) (Karlen, Andrews, & Doran, 2001). Soil health refers more specifically to the soil's semidynamic properties that can "change as a result of soil use and management over the human time scale" (Moebius-Clune et al., 2016). Given this subtle difference, we agree with a recent review that suggested "soil health" should replace the term "soil quality," and interpretations of soil health evaluations should be made relative to the constraints placed by inherent properties (Bunemann et al., 2018).

Indeed, a widely used soil health assessment protocol has identified the need to score soils differently based on difficult-to-modify properties. The Cornell Framework's Comprehensive Assessment of Soil Health uses different scoring functions for coarse-, medium-, and fine-textured soils for a number of its diagnostic tests (Moebius-Clune et al., 2016). An analysis of the overall soil health database developed by this testing framework showed justification for regionalization of scoring functions among northeastern, midwestern, and southeastern regions of the United States (Fine, van Es, & Schindelbeck, 2017). Soil health research such as this has not yet been conducted in the western United States.

In spite of these advances in contextualizing soil health tests using texture, a recognized yet still largely unaddressed research need is to further inform soil health diagnoses from a broader, pedological and geographic perspective (Norfleet,

Core Ideas

- Soil health research needs to develop diagnostics tailored to regional groupings of soils.
- Cluster analysis of a soil survey database revealed groups distinct from U.S. Soil Taxonomy.
- Inherent structure in soil survey data serves as a means to guide soil health investigations.
- Seven soil health regions were identified in California.
- Soil survey data classification framework shows promise for place-based soil health approaches.

Ditzler, Puckett, Grossman, & Shaw, 2003). Specifically, no research that we are aware of has attempted to classify the landscape based on soil properties into groups specifically to help inform soil health diagnoses and best management practices, although several recent meta-analyses have attempted to include analyses of soil health management outcomes by USDA-NRCS Soil Taxonomy orders (Nunes, Karlen, et al., 2020) or broad U.S. geographic regions (Fine et al., 2017; Nunes, van Es, Veum, Amsili, & Karlen, 2020). Such research could help guide policy to improve soil health in a more place-based logic, optimizing use of resources towards this end.

In 2019, the California legislature appropriated US\$28 million for the state's "Healthy Soils" program to award grants to farmers of up to \$100,000 for a suite of management practices such as applying compost and cover cropping, with soil organic carbon (SOC) measurement pre-project and 1, 2, and 3 yr after implementation (CDFA, 2020). As public policy expands to influence soil health management, including broader societal goals such as offsetting greenhouse gas emissions through soil carbon sequestration (Minasny et al., 2017), how soil health indices and outcomes should be evaluated across diverse landscapes and crops is a substantial challenge for soil science and farmers (Andrews, Flora, Mitchell, & Karlen, 2003). For example, there is recent evidence that soil health practices like cover cropping have differential effects on different textured soils (Jian, Du, Reiter, & Stewart, 2020).

This research seeks to address these research and policy challenges by developing a conceptual soil health region model for a major U.S. agricultural region, the Central Valley and Central Coastal Valleys of California, producing a regional grouping of soils that share similar suites of inherent soil properties, including both relatively unchangeable and semidynamic soil properties reflective of unique environmental conditions encompassing different soil systems. The goal is to repackage the USDA-NRCS Soil Survey Geographic (SSURGO) database into a map of "soil health identity" in California that reflect trends in soil performance limitations

and likely differential responses of soil landscapes to soil health management practices. We test the hypothesis that (a) a spatial coherence in SSURGO near-surface (0–30 cm) soil properties and depth to restrictive horizons should be viewable from the regional scale in California, hence systematic soil variability exists that influences the diagnosis and management of soil health, and (b) that soil health regions will be distinguishable from USDA-NRCS Soil Taxonomy orders (Soil Survey Staff, 1999) and offer a more meaningful classification of soil performance relevant to agricultural soil health diagnoses.

2 | MATERIALS AND METHODS

2.1 | Environmental setting

Given the Central Coastal and Central Valleys' internationally important agricultural production in a state grossing over \$50 billion in revenue per year (CDFA & NASS, 2018), their diverse soils make an ideal testbed for determining whether or not a complex region can be classified to better inform soil health diagnoses and policy. The Central Valley (including the Sacramento, San Joaquin, and Tulare basin) began filling with sediment approximately 150 million yr ago to a phenomenal thickness of 13–15 km from both the uplifting Coast Range on the Central Valley's west side and Sierra Nevada and Cascades on the Central Valley's east side (Graham & O'Geen, 2016). With the Central Coastal Valleys' (including the Napa, Pajaro, Salinas, Santa Clara, and Sonoma Valleys) cooler climate and mixed alluvium, the overall study area contains a broad span of age, geologic materials, climate, and topography. This covers, for example, recently deposited granitic material from glacial outwash events in the southern Sierra Nevada to older alluvial terraces with pronounced weathering and development of root restrictive horizons, such as duripans and claypans, to older soils developed in relatively dry, hot climates with pronounced alkaline and salt-affected chemistry to large swaths of shrink–swell landscapes.

2.2 | SSURGO data preparation

Several steps were needed to prepare SSURGO data for cluster analysis to identify conceptual agricultural soil health regions. First, the 2018 SSURGO shapefile of soil map units in California was clipped to the Central Valley and Central California Coastal Valleys, using the Major Land Resource Area shapefile. This selected an area of interest with 5,614,257 ha, containing 5,043 unique soil map unit keys and 3,942 unique map unit names. In SSURGO, the map unit is the basic unit of a soil map. Map units are differentiated by differences in soil types (components) and/or by management considerations such as slope. Map units consist of one or more major components, and lesser

degrees of minor components depending on the degree of soil variability and the intended use of the soil survey.

Horizon level soil properties were aggregated to 0-to-30-cm depth weighted averages or, in the case of available water capacity (AWC) and SOC content, sums by soil component. The SOC content was calculated from SSURGO as follows:

$$\text{SOC (kg m}^{-2}\text{)} = \frac{\text{OM}}{1.72} \times \text{BD} \times (1 - \text{CF}) \times \frac{Z}{10} \quad (1)$$

where OM = organic matter (%), BD = bulk density at 1/3 bar (in g cm⁻³ fines), CF = coarse fragments volumetric fraction (>2 mm), Z = horizon thickness (cm), and 1.72 is the van Bemmelen factor that assumes OM is 58% carbon by mass.

Data aggregation was done for each major soil component (defined as ≥15% areal coverage within its respective map unit) in the area of interest using the *slice* and *profileApply* functions from the *aqp* package (Beaudette, Roudier, & O'Geen, 2013) and some custom functions developed in R software (R Core Team, 2018). Component percent-weighted averages of 0-to-30-cm soil properties were then calculated for each map unit within the area of interest. Depth to restrictive horizons for each major soil component was used to calculate a component percent-weighted average of minimum depth to a restrictive horizon by type: lithic contacts, paralithic contacts, duripans, abrupt textural changes, natric horizons, saline horizons, strongly contrasting textural stratification, densic materials, petrocalcic layers, and cemented horizons. Then, the minimum component percent-weighted depth to a restrictive layer within a map unit was selected as a numerical proxy for rooting limitations. If a map unit had no reported restrictive horizons, then 200 cm was the assumed depth to a restrictive layer. If a soil component had a reported value of 0 for OM, cation exchange capacity (CEC), saturated hydraulic conductivity (K_s), or AWC, the value was assumed to be unreliable and not used. If a soil map unit had missing electrical conductivity (EC) data, salinity was assumed to be negligible and a value of 0 was assumed. No properties for rock outcrop as a major component in soil map units were included in the analysis. An either–or criterion was then used to decide if a soil map unit was included in the analysis: (a) data available for at least 80% of the map unit components or (b) data availability at least equal to the dominant component percentage. This reduced the study area to 5,274,720 ha (94% of original area of interest) and 4,595 unique map unit keys: 3,607 map units had one major component; 799 had two major components; 186 had three major components; and three had four major components.

2.3 | Data standardization

Nine 0-to-30-cm soil properties (clay, OM, CEC, linear extensibility percentage [LEP], pH, EC, BD, K_s , and AWC) and the

TABLE 1 Pearson correlations (r) among transformed and scaled soil properties in the USDA-NRCS SSURGO (Soil Survey Geographic) database. Each unique map unit key in the study area was included once in the analysis ($n = 4,595$)

Property	Restr.	Clay	OM	CEC	BD	EC	pH	LEP	Ks	AWC
Restr.	1									
Clay	-.04	1								
OM	.01	.31	1							
CEC	-.06	.91	.44	1						
BD	.07	-.70	-.45	-.71	1					
EC	.14	.08	-.12	.04	-.02	1				
pH	.30	.15	-.18	.11	.00	.56	1			
LEP	.04	.92	.23	.83	-.61	.10	.22	1		
Ks	.03	-.91	-.25	-.85	.64	-.17	-.18	-.83	1	
AWC	.01	.41	.36	.50	-.43	-.26	-.05	.37	-.43	1

Note. Restr., depth to restrictive layer; OM, organic matter; CEC, cation exchange capacity; BD, bulk density; EC, electrical conductivity; LEP, linear extensibility percentage; K_s , saturated hydraulic conductivity; AWC, available water capacity.

minimum depth to a restrictive layer (see above) were selected to represent a broad suite of physical and chemical soil properties. A number of these properties were strongly correlated to clay content (Table 1), but it was decided to use a broad suite of both invariant and semidynamic properties often included in soil health diagnostics (Moebius-Clune et al., 2016) for the purpose of this study's cluster analysis (see Section 2.4). In order to weight these properties equally in the analysis and de-emphasize extreme values, some data were log transformed and all data were then rescaled to produce standardized distributions with means equal to 0 and standard deviations equal to 1. If an EC value was less than 1.5, this was regarded as practically negligible and the value was assumed to be 0, so as to numerically disregard differences in the very low-to-low EC range. Some of the soil properties were first log transformed using the following equation:

$$\log_{10}(a + b) \quad (2)$$

where a = OM (%), CEC (cmol_c kg⁻¹ soil), EC (dS m⁻¹), LEP (%), or K_s (μm H₂O s⁻¹), and b = 0, 5, 2, 5, or 0 for each of these soil properties, respectively. All data were then standardized by subtracting the mean and dividing by the standard deviation.

2.4 | K-means cluster analysis and visualization of soil health region data

K-means clustering of transformed and standardized data was used to generate soil health region conceptual models and was performed using the *kmeans* function from the *stats* package in R (R Core Team, 2018). Since the number of clusters must be defined by the analyst using the K-means algorithm, different cluster sizes were tested (from 2 up to a maximum of 20

clusters), where each data cluster is regarded as a conceptual soil health region. The rate at which the total within-cluster sum of squares decreased as a function of increasing cluster size was used as a metric to evaluate clustering options, whereby a kink in the curve signals the optimal cluster size (Hastie, Tibshirani, & Friedman, 2009). Additionally, several other metrics were used to help identify the optimal cluster size: the average silhouette width criterion (Rousseeuw, 1987) and Calinski-Harabasz index (Caliński & Harabasz, 1974) from the *kmeansrun* function in the *fpc* package in R (Hennig, 2019) and the "gap" statistic (Tibshirani, Walther, & Hastie, 2001) from the *clusGap* function in the *cluster* package in R (Maechler, Rousseeuw, Struyf, Hubert, & Hornik, 2019).

Visualization of cluster size options was accomplished by plotting the standardized means of each soil property by cluster (soil health region) in a spider plot using the *radarchart* function from the *fmsb* package in R (Nakazawa, 2018). Then, for each cluster size, soil property data distributions were compared in side-by-side violin plots to visualize differences in properties amongst regions, using the *vioplot* function from the *vioplot* package in R (Adler & Kelly, 2019).

2.5 | Validation of soil health region conceptual model and comparison with Soil Taxonomy

Two point datasets were used to assess the validity of cluster size options. Three hundred and sixty-nine sampling locations within the study area from the Kellogg Soil Survey Laboratory (KSSL locations) were used to assess group mean statistical differences in nine 0-to-30-cm soil properties across the soil health regions (NCSS, 2019). An additional 76 sampling locations from 16 vineyards in Napa Valley and 27 points from five vineyards in the Lodi Winegrape district in the Central

Valley (Napa-Lodi locations) were included as part of the validation exercise (Burns et al., 2015, 2016; Yu et al., 2017, 2019). These 103 Napa-Lodi locations had data for four of the nine evaluated KSSL soil properties. Soil health region identity was assigned to each location based on which particular region the sampling point was located across cluster sizes from three- up to the nine-region cluster sizes (each can be regarded as a conceptual model option). Then, differences in mean soil properties among soil health regions for each conceptual model were assessed by an overall one-way ANOVA followed by Tukey's honest significant difference test ($\alpha = .05$) for the following 0-to-30-cm soil data: clay ($n = 314$), OM ($n = 414$), SOC stocks ($n = 209$), BD ($n = 214$), pH ($n = 350$), EC ($n = 151$), CEC ($n = 233$), LEP ($n = 124$), and AWC ($n = 124$). Many of the KSSL sampling points were likely used to develop SSURGO map unit concepts and guide population of the SSURGO database, so they do not represent an entirely independent validation of the *K*-means clustering of SSURGO data. However, in addition to the 103 Napa-Lodi locations, at least 54 of the 369 KSSL sampling locations are known to be sourced from the California Soil and Vegetation Survey not used to populate the SSURGO database, all located in the northern half of the study area.

Finally, the same SSURGO soil properties used in the cluster analysis were summarized by Soil Taxonomy at the order level as a comparison to results from *K*-means unsupervised classification. Each map unit had its dominant soil order identified by summation of the major component percentages by soil order. In addition, KSSL and Napa-Lodi sampling locations were assigned a soil order based on the SSURGO database. This allowed for the same one-way ANOVA of validation locations by Soil Taxonomy orders.

3 | RESULTS

3.1 | Selection of a soil health region conceptual model

The development and selection of a soil health region conceptual model required SSURGO training data aggregation (see Section 2.2) and standardization (see Section 2.3), *K*-means cluster analysis of 10 selected properties from the standardized SSURGO data to develop options of different possible cluster sizes (see Section 2.4), and concluded with evaluation of the resulting conceptual model options using various approaches compared to Soil Taxonomy orders (see Section 2.5) as follows.

First, the clustering metrics highlighted that a range of cluster sizes for a soil health region conceptual model were justifiable, since the metrics did not point to a specific number of clusters as unequivocally the best. Several kinks in a plot of total within-cluster variance by cluster size suggested that

there was not a unique identifiable cluster size using SSURGO data (Figure 1). Other metrics were contradictory. The gap statistic increased through twenty data clusters (the largest cluster size evaluated), suggesting that increased grouping of the data was useful. However, both the average silhouette width and Calinski–Harabasz index declined from a peak at two data clusters, suggesting less partitioning of the data was better.

Second, the validation exercise using KSSL and Napa-Lodi point data organized by cluster assignment (for each cluster size, each validation point had a cluster label assigned based on its geographic location) showed a number of statistically significant group contrasts across the cluster size options for most soil properties (Figure 2). Clay, pH, CEC, OM, and LEP showed 50% or more possible group mean contrasts as significant. In contrast, EC, BD, and AWC had five, two, and zero significant contrasts of 21 possible group contrasts for the seven-region model (Figure 2). To put this in context, given a statistical significance threshold of 5% probability, there would only be one expected group contrast out of 21 possible group contrasts for a random seven-region model. While the three- and four-region models, both had the highest possible proportion of significant group mean contrasts across the eight soil properties, the total number of group contrasts increased steadily through the seven-region model before leveling off through the nine-region model except for OM (Figure 2). Specifically, compared with the six-region model, the seven-region model added four to six significant group contrasts for pH, clay, CEC, LEP, and EC and two significant group contrasts for OM out of the six additional possible group contrasts, whereas the eight-region model added another two to three significant group contrasts for clay, OM, pH, and CEC and one significant group contrast for BD and EC out of the seven additional possible group contrasts. The nine-region model added four significant group contrasts for OM with no additional group contrasts for any other properties out of the eight additional possible group contrasts compared with the eight-region model. Thus, in terms of the available validation dataset, this suggested the seven-region model as the most defensible as a possible framework for interpreting soil health diagnostics and guiding recommended practices in California's major agricultural valleys, and it is described in detail below. Some additional justifications for the seven-region model are provided in Section 4 compared with the other cluster size options.

3.2 | Descriptions of soil health regions in the seven-region conceptual model

Broadly speaking, the study area can be conceptualized as occurring in two general phases, soils with and without performance limitations to crop production (Figure 3). Soils without

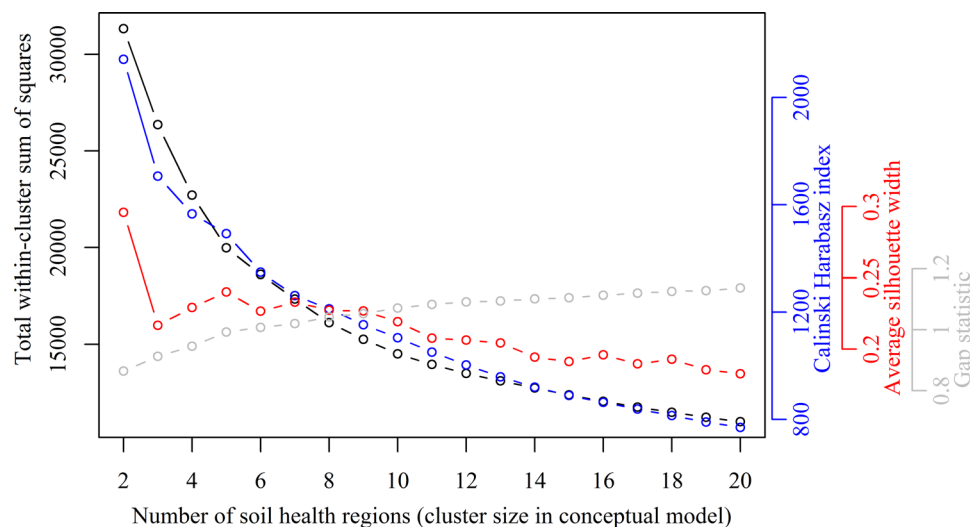


FIGURE 1 Cluster size options for the soil health region conceptual model. Various clustering metrics (y axis) according to different cluster sizes (x axis) do not suggest an optimal number of clusters. For the Calinski–Harabasz index (blue), average silhouette width (red), and gap statistic (gray), more is better. For total within-cluster sum of squares (black), less is better

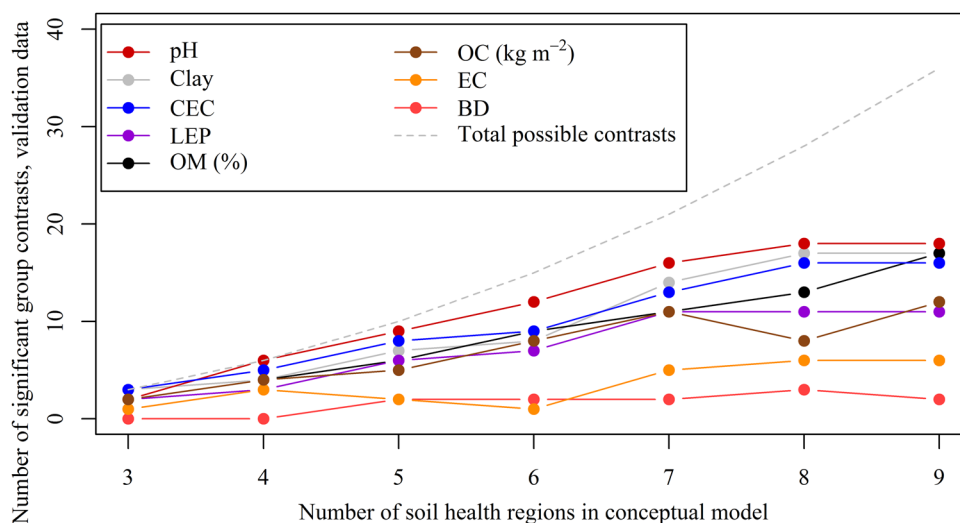


FIGURE 2 For cluster sizes of 3–9, ANOVA by soil health region for combined Kellogg Soil Survey Laboratory (KSSL) and Napa-Lodi point data was performed for the following 0-to-30-cm soil properties: pH, clay, cation exchange capacity (CEC), linear extensibility percentage (LEP), organic matter concentration (OM, %), organic carbon stocks (OC, kg m^{-2}), electrical conductivity (EC), and bulk density (BD). Analysis of variance was followed by testing for significant group contrasts in mean soil property validation data among soil health regions using Tukey's honest significant difference ($\alpha = .05$)

performance limitations (45% of study area) can be thought of as those soils with relatively homogenous profiles to at least a depth of 150 cm with no appreciable soil chemistry barriers to plant growth or physical obstacles to root elongation. As clusters, soil landscapes with performance limitations to crop production (in 55% of the study area) tend to occur as three types: (a) soils with root restrictive horizons that develop on stable landforms in both seasonally wet (e.g., duripans, claypans) and, to a lesser extent, relatively dry climates (e.g., natric and salic horizons) or in eroding, upland landscapes on valley margins with shallow depth to weathered bedrock (24%

of the study area); (b) alkaline soils with potential aggregation issues due to high Na^+ , also having propensity for problematic salinity that affects relative water availability and crop health (16% of the study area); and (c) shrink–swell soils formed from alluvium that are prone to water-logging, are inherently challenging to cultivate, and are not ideal for many crops (16% of the study area). The salt-affected region, soils with restrictive horizons, and soils without performance limitations can be further divided by texture and related properties and/or by OM levels. The overall result is a conceptual model of seven distinct soil health regions with various levels of regional

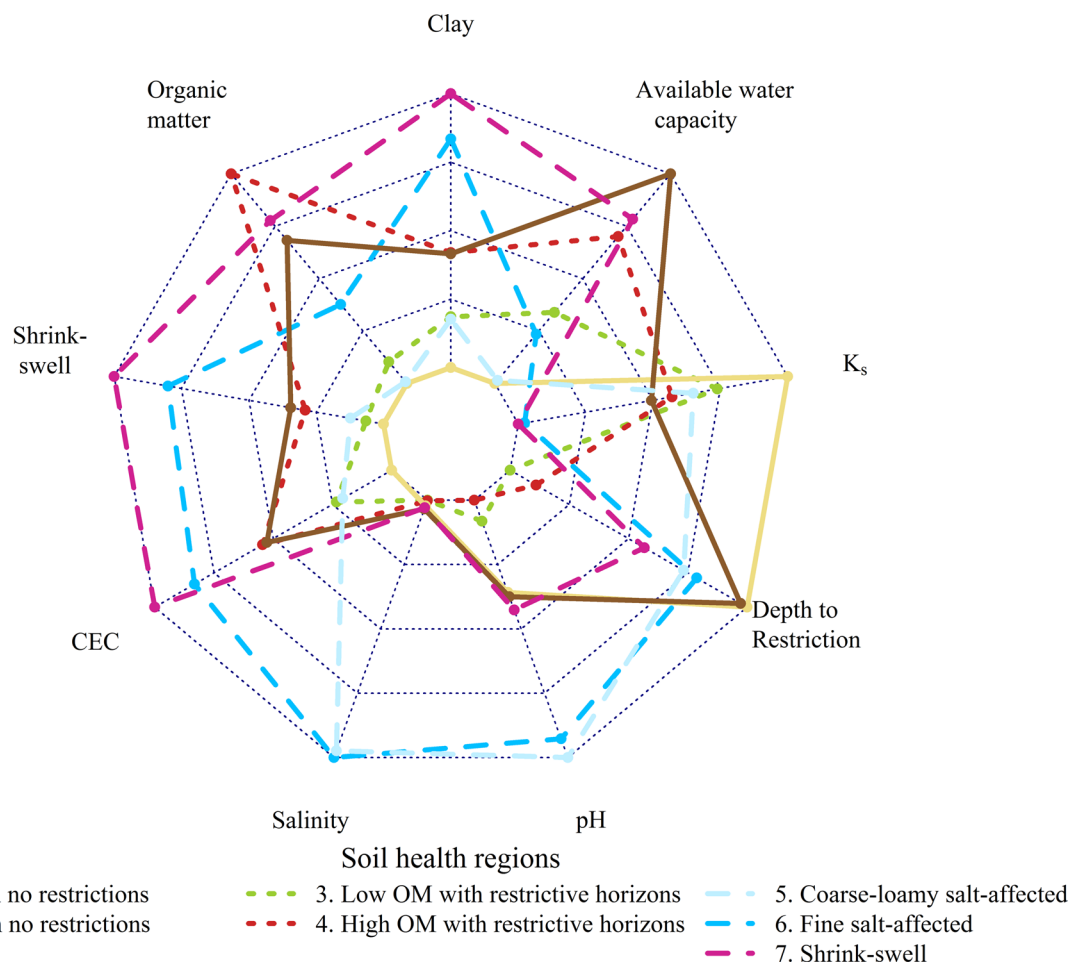


FIGURE 3 Spider chart (also known as a radarchart) showing mean soil properties (transformed, if necessary, and all properties rescaled to mean = 0 and standard deviation = 1) for seven soil health regions identified from *K*-means cluster analysis. Regions 3–7 have soil performance limitations. The outermost region of the plot are relatively high values. The innermost region of the plot are relatively low values. CEC is cation exchange capacity; OM is organic matter

separation (Figure 4) across three general soil property groups: (a) those properties generally related to clay content and mineralogy (clay strongly correlated to CEC, LEP, K_s , and BD and more weakly to OM and AWC); (b) chemistry (pH and EC); and (c) depth to restrictive layer (Table 1, Figures 5–6). In the descriptions below, representative soil series are highlighted that cover at least 30,000 ha of a particular soil health region and with at least 80% of the soil series' study area coverage assigned to only one soil health region. The following soil health regions were delineated with spatially weighted mean 0-to-30-cm soil properties noted also in parentheses.

Region 1 (coarse with no restrictions) occupies 21% of the study area and tends to be very deep, slightly acid to slightly alkaline, highly permeable, well drained, and relatively homogenous soils formed from more recent alluvial deposits. These soil landscapes have had little time for development of root restrictive horizons. They have relatively low clay content (11%), OM (0.8%), CEC (8 cmol_c kg⁻¹ soil), and

AWC (0.11 cm H₂O cm⁻¹ soil) in the surface layer (Figures 3 and 5). In the sandiest soils of this group, these can result in a performance limitation to rainfed crop production. This widespread region is typified by the Hanford, Delhi, Wasco, Hesperia, Milham, and Tujunga soil series, all Entisols except Milham, an Aridisols. The typical soil in this class is formed from relatively coarse-textured materials in alluvial fans on the eastside of the San Joaquin Valley deposited from the higher-elevation, southern Sierra Nevada granitic batholith, where glacial outwash events have been more pronounced (Figure 4). It is also found in northern valley locations closer to channels where higher energy flood events would tend to deposit coarser materials.

Region 2 (loamy with no restrictions) occupies 23% of the study area and tends to be very deep, slightly acid to slightly alkaline, slow to moderately permeable, well drained, and relatively homogenous soils formed from alluvial deposits. These soil landscapes have moderate clay content (25%), OM (1.6%), CEC (19 cmol_c kg⁻¹ soil), and high AWC (0.17 cm

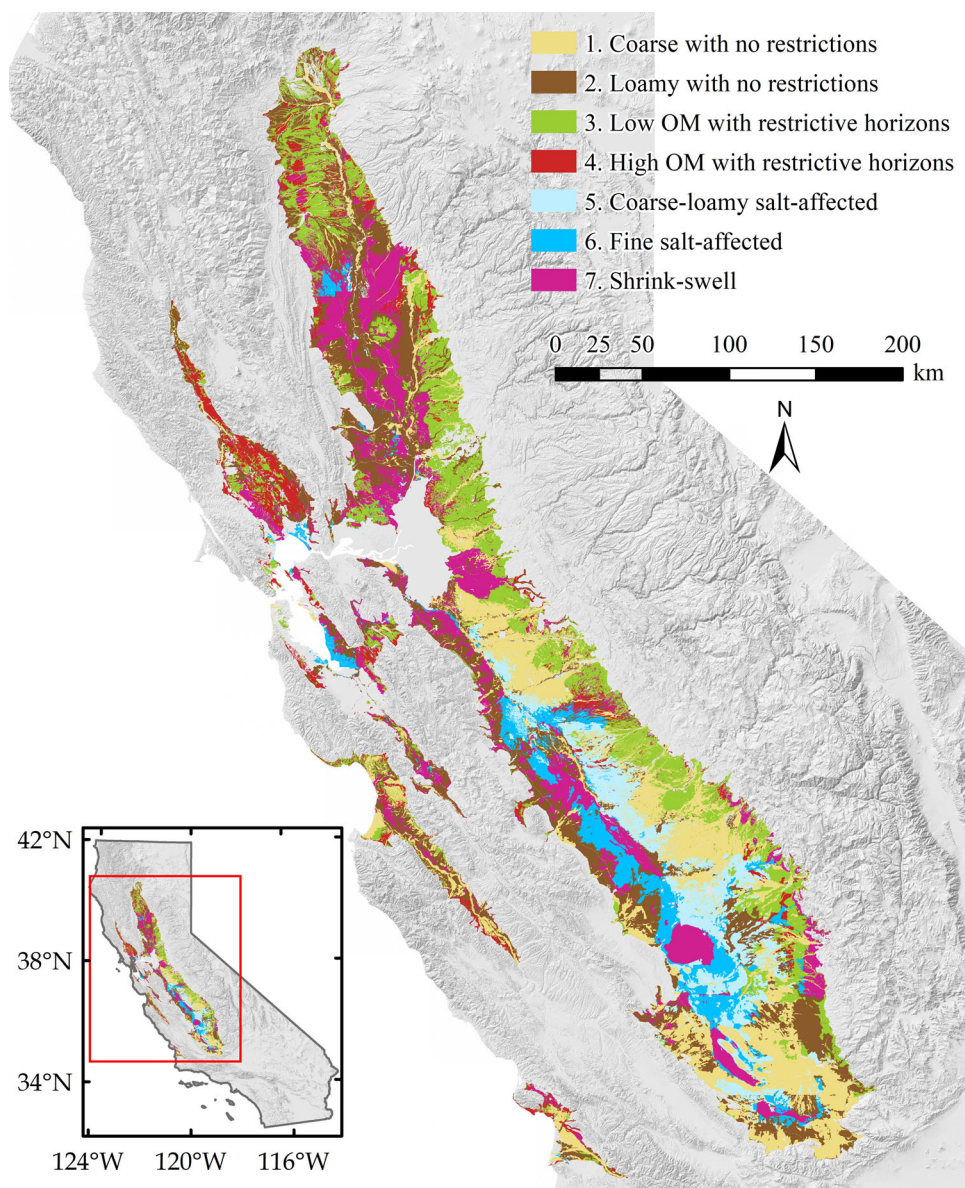


FIGURE 4 Map showing seven soil health regions identified from *K*-means cluster analysis of 10 soil properties derived from USDA-NRCS SSURGO (Soil Survey Geographic) data. Nine of the properties reflect the upper 30 cm of soil. The 10th property was the minimum depth to a root restrictive horizon. OM is organic matter

$\text{H}_2\text{O cm}^{-1}$ soil) in the surface layer (Figures 3 and 5). This widespread region of mixed soil orders (Table 2) is typified by the Panoche, Cerini, Yolo, Colpien, Chanac, and Tehama soil series. It is located primarily in the northern half of the study area where frequent low to moderate flood events have deposited finer-textured materials and on the west side Central Valley margins from finer-textured, Coast Range materials (Figure 4). The region also exists on lower terraces where there has not been enough time since alluvial deposition to develop root restrictive horizons.

Region 3 (low OM with restrictive horizons) occupies 18% of the study area and tends to be moderately deep to a restrictive layer, moderately acid to slightly alkaline, mod-

erately well drained, and formed from alluvial deposits on stream terraces or alluvial fans above current floodplains or from residuum in uplands. These soil landscapes have low to moderate clay content (18%), CEC ($12 \text{ cmol}_c \text{ kg}^{-1}$ soil), AWC ($0.13 \text{ cm H}_2\text{O cm}^{-1}$ soil), and low OM (1.0%) in the surface layer (Figures 3 and 5). Permeability can be slow as a result of the restrictive layer. This widespread region (Figure 4) is typified by the San Joaquin, Redding, Exeter, Corning, Cometa, and Madera soil series, all Alfisols with duripans or abrupt textural changes. They are most widespread on the Central Valley's eastern margins on terraces formed from granitic deposits from the eroding Sierra Nevada batholith.

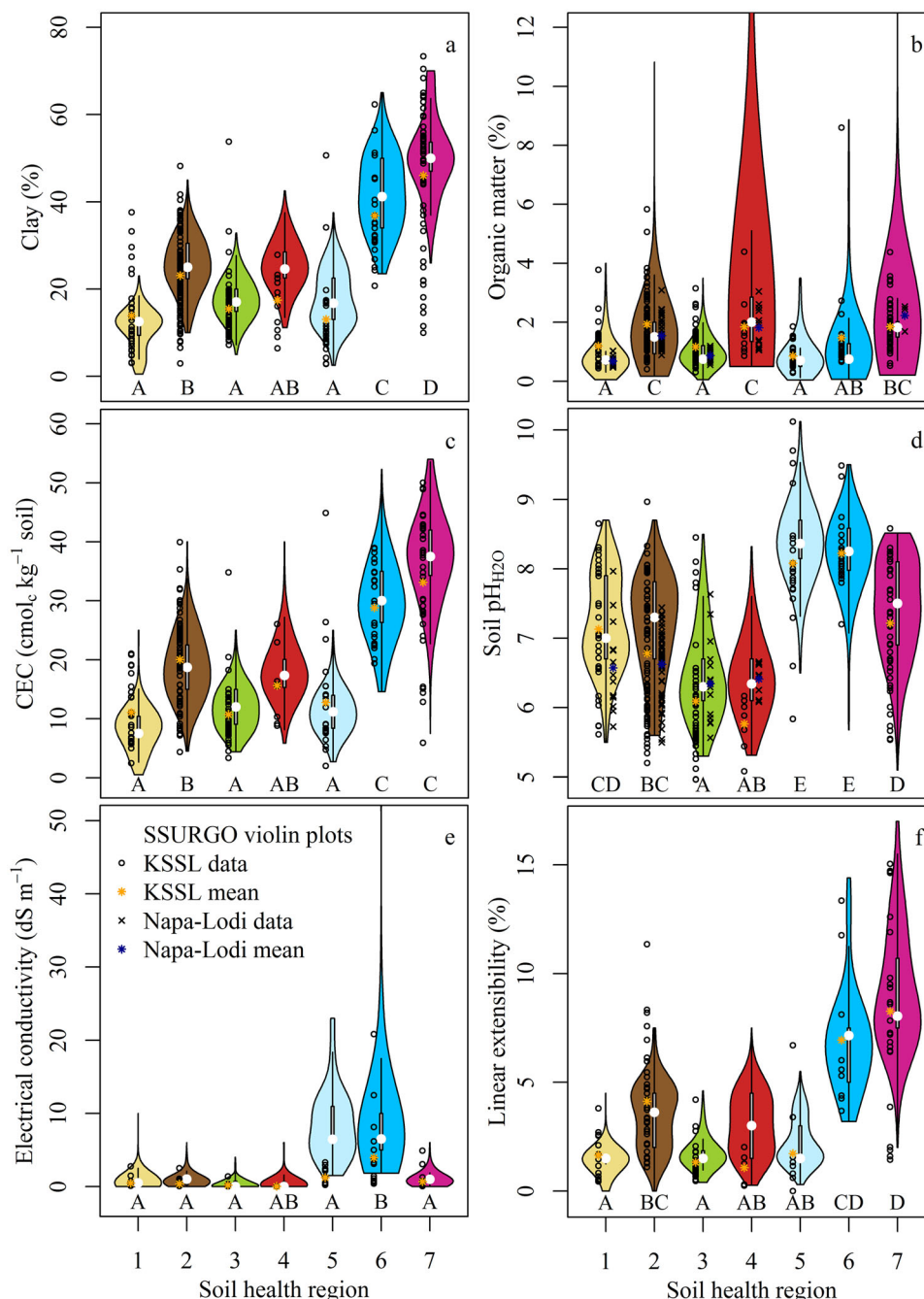


FIGURE 5 (a–f) Areal-weighted violin plots of USDA-NRCS SSURGO (Soil Survey Geographic) data for six of the nine 0-to-30-cm soil properties considered in the seven-class *K*-means cluster analysis to identify soil health regions. Colors and numbers match the legends for Figures 3–4. When available, Kellogg Soil Survey Laboratory (KSSL) and Napa-Lodi point data for 0-to-30-cm soil properties in a given region are shown alongside violin plots as a validation test of the seven-region conceptual model. Using this validation dataset, different letters denote a statistically significant difference ($\alpha = .05$) in mean soil properties among soil health regions. CEC is cation exchange capacity

Region 4 (high OM with restrictive horizons) occupies 5% of the study area and tends to be moderately deep to a restrictive layer, moderately acid to slightly alkaline, moderately well drained, and formed from alluvial deposits on stream terraces or alluvial fans above current floodplains. These soil landscapes have moderate clay content (25%), CEC (18 cmol_c kg⁻¹ soil), and AWC (0.15 cm H₂O cm⁻¹ soil) and high OM

(2.5%) in the surface layer (Figures 3 and 5). Permeability can be slow as a result of the restrictive layer. This region of mixed soil orders is typified by the Flamen soil series but is comprised of a relatively large number of soil components occurring mostly in the wetter, northern half of the study area such as shallow soils with lithic and paralithic contacts in cropped foothills near the Napa and Sonoma Valleys

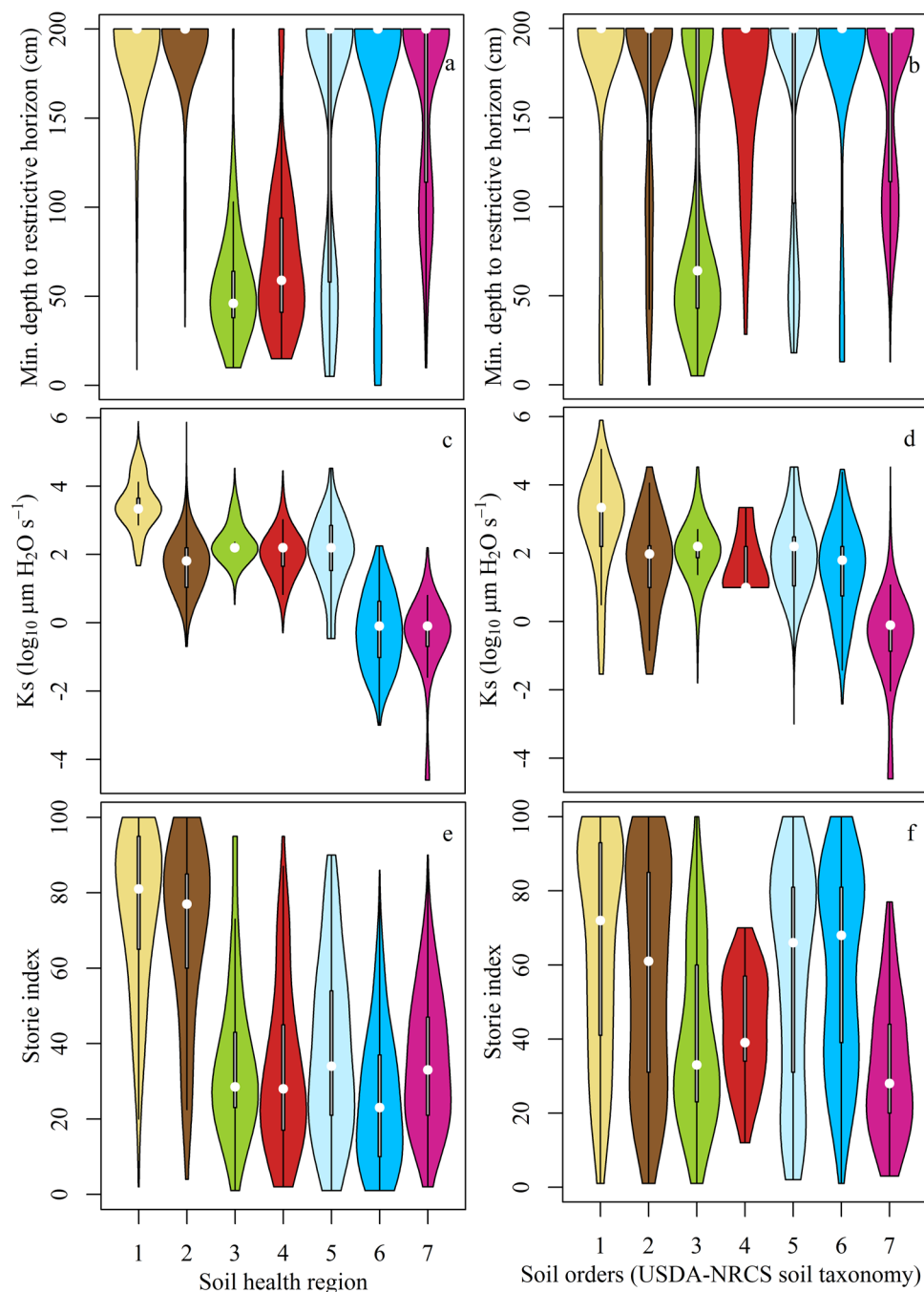


FIGURE 6 (a–f) Areal-weighted violin plots of USDA-NRCS SSURGO (Soil Survey Geographic) data for (a–b) depth to minimum (min.) restrictive horizon, (c–d) saturated hydraulic conductivity (K_s), and (e–f) the Storie Index (O’Geen et al., 2008) by both soil health region (a, c, and e) and Soil Taxonomy order (Soil Survey Staff, 1999) (b, d, and f): (1) Entisols, (2) Mollisols, (3) Alfisols, (4) Ultisols, (5) Inceptisols, (6) Aridisols, and (7) Vertisols. A map unit’s soil order was determined by sum of the major component percentage of the various soil orders and selection of the dominant soil order from that tabulation. The color of the most similar soil health region is used for each soil order (see Table 2). Productivity increases from nonagricultural (<20) to prime farmland (>80) along the Storie Index

where climate has favored higher levels of OM accumulation (Figure 4). The soils also occur on dissected high terraces, such as eastside Central Valley margins where remnant geomorphic “islands” have been cut off from more recent granitic alluvial deposits from the eroding Sierra-Nevada batholith (Graham & O’Geen, 2016).

Region 5 (coarse-loamy salt-affected) occupies 9% of the study area and is distinguished by alkaline (8.4 pH) and often-times saline chemistry (8.5 dS m^{-1}) with a substantial proportion of the cation exchange complex occupied by sodium (16 sodium adsorption ratio). These soils have low to moderate clay (18%), CEC ($11 \text{ cmol}_c \text{ kg}^{-1} \text{ soil}$), and AWC (0.11 cm

TABLE 2 Soil health regions are comprised of various Soil Taxonomy orders (Soil Survey Staff, 1999) (upper half of table) and vice versa (lower half of table). Andisols (18 ha), Ultisols (18,980 ha), and map units having more than one major component with differing, but equally dominant soil orders (71,361 ha) cover relatively small portions of the study area and are omitted from the table. Thus, rows (upper half of table) and columns (lower half of table) do not necessarily sum exactly to 100%. A map unit's soil order was determined by sum of the major component percentage of the various soil orders and selection of the dominant soil order from that tabulation

Region	Alfisols	Aridisols	Entisols	Inceptisols	Mollisols	Vertisols	Total
	%						ha
Soil health region by soil orders							
1. Coarse with no restrictions	13	9	62	2	14	0	1,133,584
2. Loamy with no restrictions	22	14	11	15	36	2	1,232,173
3. Low OM ^a with restrictive horizons	80	1	6	2	10	0	964,182
4. High OM with restrictive horizons	32	0	3	18	39	0	285,112
5. Coarse-loamy salt-affected	24	12	28	14	15	0	469,939
6. Fine salt-affected	13	31	22	4	14	16	367,662
7. Shrink–swell	2	2	6	0	20	69	822,067
Soil order by soil health regions							
1. Coarse with no restrictions	10	22	61	6	14	0	
2. Loamy with no restrictions	18	36	12	50	41	3	
3. Low OM with restrictive horizons	53	2	5	6	9	0	
4. High OM with restrictive horizons	6	0	1	14	10	0	
5. Coarse-loamy salt-affected	8	12	11	18	6	0	
6. Fine salt-affected	3	24	7	4	5	9	
7. Shrink–swell	1	4	4	1	15	88	
Total, ha	1,456,896	469,453	1,157,302	364,166	1,094,093	642,451	5,274,720

^aOM, organic matter.

TABLE 3 Spatially weighted means and interquartile range (in parentheses) of climate properties by soil health region

Region	Annual <i>P</i>	Annual <i>ET</i> _o	Annual <i>T</i>	Aridity index
	mm		°C	
1. Coarse with no restrictions	309 (147)	1,428 (97)	17.0 (1.1)	0.22 (0.12)
2. Loamy with no restrictions	419 (320)	1,368 (152)	16.7 (0.7)	0.32 (0.26)
3. Low OM ^a with restrictive horizons	475 (251)	1,384 (82)	16.7 (0.6)	0.35 (0.20)
4. High OM with restrictive horizons	668 (472)	1,252 (231)	15.9 (1.8)	0.56 (0.47)
5. Coarse-loamy salt-affected	234 (69)	1,462 (32)	17.2 (0.6)	0.16 (0.05)
6. Fine salt-affected	253 (92)	1,430 (73)	17.1 (0.7)	0.18 (0.07)
7. Shrink–swell	409 (259)	1,369 (106)	16.7 (0.5)	0.31 (0.21)

Note. *P*, precipitation; *ET*_o, reference evapotranspiration; *T*, temperature; aridity index = *P*/*ET*_o.

^aOM, organic matter.

H₂O cm^{−1} soil) and low OM (0.8%) (Figures 3 and 5). Soils also often have restrictive horizons (duripans, claypans, and natric) that result in slow or very slow permeability, in spite of their relatively low clay content. This region of mixed soil orders (Table 2) is typified by the Fresno and Garces series and occurs almost exclusively in the drier, southern half of the study area along the San Joaquin River (Figure 4, Table 3).

Region 6 (fine salt-affected) occupies 7% of the study area and is distinguished by alkaline (8.2 pH), oftentimes saline (8.4 dS m^{−1}) chemistry with a substantial proportion of the cation exchange complex occupied by sodium (18 sodium adsorption ratio). These soils have high clay content (42%), typically with high shrink–swell capacity (6.9 LEP). These soils also have relatively low OM (1.2%) and AWC (0.12 cm H₂O cm^{−1} soil) but high CEC (30 cmol_c kg^{−1} soil)

(Figures 3 and 5) and low permeability (Figure 6c). Root restrictive horizons (duripans, claypans, and natric) are common. This region of mixed soil orders (Table 2) is typified by the Calflax and Lethent soils series and occurs mostly in the southern half of the study area with some notable exceptions where saline, Coast Range sourced alluvium or dry climate and poor drainage conditions have led to similar properties (Figure 4, Table 3).

Region 7 (shrink–swell) occupies 16% of the study area and are the most finely textured soils in the landscape with shrink–swell clays (9.2 LEP, 51% clay) but without pronounced alkalinity (7.4 pH) or salt-affected chemistry. They have high AWC (0.15 cm H₂O cm⁻¹ soil) and high CEC (38 cmol_c kg⁻¹ soil) and hold a substantial amount of organic carbon (1.9% OM) (Figures 3 and 5), but have very low permeability (Figure 6c). This widespread region occurs throughout the Central and Coastal Valleys in alluvium deposited in low-energy floods (Figure 4) and is typified by the Capay, Clear Lake, Willows, and Tulare soil series, with the region almost all either Vertisols (69%) or Mollisols (19%).

3.3 | Comparing the soil health region conceptual model with Soil Taxonomy

Another path to a conceptual soil health region model would be to use the existing U.S. soil classification system to group the landscape into regions. Eight of the twelve soil orders from Soil Taxonomy were represented in the study area and also showed general geographic coherence (Supplemental Figure S1). Three soil orders (Alfisols, Entisols, and Mollisols) comprised 70% of the landscape; three less extensive soil orders (Aridisols, Inceptisols, and Vertisols) comprised another 28% of the landscape (Table 2). Andisols and Ultisols together represent <2% of the study area and are omitted from Table 2. Several soil orders corresponded relatively well with the conceptual soil health regions. Sixty-two percent of Region 1 (coarse with no restrictions) are Entisols, Eighty percent of Region 3 (low OM with restrictive horizons) are Alfisols, and 69% of Region 7 (shrink–swell) are Vertisols. However, Alfisols, the most dominant soil order in the study area comprising 28% of the overall landscape, covered >20% of various soil health regions. Alfisols comprised 22% of Region 2 (loamy with no restrictions), 33% of Region 4 (high OM with restrictive horizons), and 24% of Region 5 (coarse-loamy salt-affected). Moreover, in four regions, the most dominant soil order did not comprise an overall majority: 36% of Region 2 (loamy with no restrictions) are Mollisols; 37% of Region 4 (high OM with restrictive horizons) are Mollisols; 28% of Region 5 (coarse-loamy salt-affected) are Entisols; and 31% of Region 6 (fine salt-affected) are Aridisols. Indeed, similar to Alfisols, >10% of both Mollisols and Aridisols were dispersed among four dif-

ferent soil health regions. At the other end of the Soil Taxonomy spectrum, using what is comparable to the soil series concept in SSURGO data, there was a total of 849 unique major component names in the study area with a range of 75–291 (mean = 184) unique major component names by soil health region.

Assigning dominant soil orders to the KSSL and Napa-Lodi locations based on information from the SSURGO database allowed for another validation exercise, this time comparing the seven-region soil health conceptual model with the highest level of Soil Taxonomy (Supplemental Figure S2). In terms of significant group contrasts, the seven-region model developed from unsupervised classification outperformed Soil Taxonomy for clay (14 vs. 10 significant contrasts), CEC (13 vs. 10 significant contrasts), LEP (11 vs. 9 significant contrasts), EC (5 vs. 0 significant contrasts), and SOC content (11 vs. 5), but not OM (both had 11 significant contrasts), pH (both had 16 significant contrasts), or AWC (both had no significant contrasts). However, the Napa-Lodi and KSSL locations identified as Ultisols, which comprise <1% of the study area, disproportionately accounted for 5 of 16 significant group contrasts across soils orders for pH and 4 of 11 significant group contrasts for OM. Moreover, the unsupervised classification led to a better partitioning of the data according to agronomic performance limitations compared to Soil Taxonomy, as captured by the Storie Index (an agricultural productivity rating developed for California) (Figure 6e–f).

4 | DISCUSSION

4.1 | Development and validation of a soil health region model

The seven-region soil health conceptual model selected in this study presents a geographically coherent snapshot of some general California soil health challenges in its most agriculturally important domain, the Central and Coastal Valleys. From a general perspective, the unsupervised classification revealed a four-group structure from the complex, many-to-one SSURGO database, distilling 10 selected soil properties and thousands of uniquely named soil components into a primary group with relatively homogenous soil profiles and no notable performance limitations and three additional groups with differing soil performance limitations to crop production: (a) root restrictive horizons; (b) strongly alkaline and salt-affected; and (c) excessive shrink–swell (Figure 3). From this four-group structure, soils with homogenous profiles and no performance limitations, soils with restrictive horizons, and salt-affected soils were effectively subdivided into regions with either contrasting soil texture, the property linked most strongly to other soil properties (Table 1), and/or OM levels (Figures 3 and 5). This produced a total of seven

TABLE 4 Percentage of irrigated crop classes and managed wetland by soil health region identified by 2014 remote sensing (CDWR, 2017), which classified 64% of the 5,274,720 ha study area. Several land classes identified by remote sensing are not included in the table: 12,565 ha of young perennials and 8,936 ha of specialty crops

Crop/land use	No performance limitations									Performance limitations	Hectares
	1	2	3	4	5	6	7				
	%									ha	
Alfalfa & pasture	15	24	38	13	3	17	11	19	62	273,819	
Annual field crops	11	24	34	8	2	13	11	30	66	739,213	
Grapes	43	20	63	16	4	11	1	4	37	291,216	
Orchard	31	28	58	17	2	10	3	10	42	932,420	
Small fruits & vegetables	30	42	72	3	1	3	4	15	28	167,589	
Managed wetland	7	11	17	4	2	18	30	29	83	156,407	
Urban	30	24	54	25	5	2	1	14	46	423,760	
Idle	10	24	35	10	2	6	19	29	65	375,213	

Note. 1 = coarse with no restrictions; 2 = loamy with no restrictions; 3 = low organic matter with restrictive horizons; 4 = high organic matter with restrictive horizons; 5 = coarse-loamy salt-affected; 6 = fine salt-affected; 7 = shrink-swell.

conceptual soil health regions with differing sets of properties (Figures 5 and 6). This classification effort is unique in presenting a novel perspective of how groups of soil properties tend to cluster at regional scale, revealing a natural differentiation of the soilscape without reference to any preconceived notions of pedogenic process and Soil Taxonomy. Other soil health research, including mapping efforts, have revealed correlations between various soil properties and land use (Seaton et al., 2020; Svoray, Hassid, Atkinson, Moebius-Clune, & van Es, 2015), but these studies have not attempted to unravel the underlying correlation between relatively invariant properties and the land use decisions purported to relate to the soil properties themselves. In California, there is a high degree of correspondence among different crop groups cultivated and the soil health regions (Table 4), suggesting that crop and management choice is influenced to a large degree by inherent soil properties. Thus, retrospective studies of land use effects on soil health need to be interpreted carefully to control for this inherent connection between land management and soil properties.

Unfortunately, there was no clearly objective metric to identify the optimal number of soil health regions. For example, further division of the landscape beyond seven regions into a more complex conceptual model was not justified by a validation exercise (Figure 2), although additional SSURGO data variance could be captured at a similar rate by clustering from a five-region through a 12-region model as seen in the steadily decreasing within-cluster sum of squares (Figure 1). However, the Calinski–Harabasz index and silhouette statistic highlighted an apparent lack of high quality clusters in the data, also evident in the broad soil property ranges for each region (Figures 5 and 6), in contrast with the gap statistic, which justified parsing SSURGO into the maximum number of groups considered in the cluster analysis (20). How-

ever, in a previous simulation study of these commonly used clustering metrics, only the gap statistic could identify when uniformly distributed data did not have any cluster structure, whereas by definition both the Calinski–Harabasz index and silhouette statistic can identify no fewer than two clusters (Tibshirani et al., 2001). Thus, the gap statistic results supported that the cluster structure was real and the use of a validation dataset to guide selection of a justifiable number of regions, following a general cluster analysis recommendation from statisticians: “...results of a clustering analysis...should constitute a starting point for the development of a scientific hypothesis and further study, preferably on an independent data set” (James, Witten, Hastie, & Tibshirani, 2013).

Some specific results from this validation exercise highlighted the possible advantages of the seven-region model vs. both simpler and more complex models. The five-region model lumped Regions 3 and 4 into a general group of soils with restrictive horizons and Regions 6 and 7 into a general group of fine-textured soils with some soils from Region 6 lumped with a general region of salt-affected soils. Regions 3 and 4 (soils with restrictive horizons) were effectively separated with regards to OM, clay, and CEC in the six-region model, although the validation data only supported an effective separation by OM. Next, Regions 5, 6, and 7 were effectively separated in the seven-region model with regards to pH, texture, and EC (Figures 3 and 5) compared with the six-region model. However, the eight- and nine-region models primarily separated Region 1 (coarse with no restrictions) into additional textural variants, but these could not be statistically separated at the KSSL and Napa-Lodi validation locations. Moreover, further separation of coarse-textured groups is less relevant in California where most cropping systems are irrigated.

Given the somewhat subjective nature of deciding on the seven-region model as a framework for evaluating soil health in California, the map is intended to be tested further by datasets not currently available and is not presented as conclusive. It is certainly possible that future research and experience will suggest that some of the regions are not helpful or need to be further subdivided to demarcate mappable soil health contextual differences. It was designed with the intent of being a starting point that will evolve.

In the validation exercise, regional differences in soil properties were apparent for relatively invariant (e.g., clay) and semidynamic (e.g., OM) properties (Figure 2). This highlights OM as a property reflective of whole soil systems and their unique environmental conditions, which encompass the totality of ecosystem properties for a given location. Since OM is typically considered to be an overarching soil health indicator, this research emphasizes the importance of considering the overall environmental context of soil health analyses when evaluating management outcomes (Nunes, van Es, et al., 2020). However, the semidynamic BD and AWC, both commonly used in soil health analyses (Moebius-Clune et al., 2011), showed very few significant differences among regions in the validation dataset (Figure 2). Differences in AWC would have likely been more evident if full rooting zone sums were compared to capture the effect of root restrictive horizons.

The conceptual model could be used to test hypotheses about whether or not clustering of soils based on largely physical and chemical properties helps interpret a more biological set of soil health diagnostics like those used in the Cornell Framework's Comprehensive Assessment of Soil Health (CASH) methodology, such as active carbon, soil respiration, and soil proteins (Moebius-Clune et al., 2016). At the same time, a validation exercise demonstrates that although useful in distinguishing the mean property differences among regions, the conceptual model is far from perfect, showing overlap in point data among some regions for certain properties (Figure 5). Since validation points were assigned to a soil health region by location, this could be partly due to map-unit design, such that a KSSL or Napa-Lodi sampling location hit a minor component or inclusion within a map unit. However, this result also highlights the fact that soil exists as a continuum across the landscape (Knox, 1965) with variability often fractal in nature (Burrough, 1983), so any regional approach at classification is bound to at least some overgeneralization. Indeed, the SSURGO data show overlapping properties among regions that are distinguished by their given names (e.g., % clay comparing coarse with no restrictions vs. loamy with no restrictions) (Figure 5a), meaning that some portions of the map are not easily classified and perhaps could be better demarcated as areas of uncertainty (Figure 4).

4.2 | Relationship of the soil health region conceptual model to existing Soil Taxonomy

This use of unsupervised learning (*K*-means cluster analysis) to reveal underlying but unexploited structure in the SSURGO database is distinguished from a research goal to exploit machine learning to assign predefined taxonomic classes to soil observational datasets (Arkley, 1976). A comparison of the classification derived from *K*-means clustering with Soil Taxonomy at the order level highlighted why new approaches are likely needed to define regions for soil health research and diagnostics. Specifically, an integrated soil productivity rating called the Storie Index (O'Geen, Southard, & Southard, 2008), which was developed in California, showed better clustering by soil health region compared to organizing the data by soil order (Figure 6e–f). Since the Storie Index was not used as an input dataset for the *K*-means cluster analysis, this helps validate the proposed framework as a means to classify the landscape in terms of soil performance limitations to crop production connected to soil health diagnoses.

Although three soil health regions aligned relatively well with Soil Taxonomy orders, four of seven regions were mixtures of multiple, relatively equally represented soil orders (Table 2). A particular deficiency in Soil Taxonomy for California and other arid to semiarid regions is the lack of a well-defined group for salt-affected and alkaline soils. Although many Aridisols have these characteristics, the criteria for this soil order do not completely capture this important performance limitation and other soil health challenges largely unique to semiarid and arid climates. Large portions of the salt-affected regions are also identified as Alfisols, Mollisols, Vertisols, and Entisols (Table 2). Salinity in general is a semidynamic property affected by irrigation, climate, and soil profile characteristics that span several soil orders. The lack of correspondence between the unsupervised classification results and Soil Taxonomy is not necessarily surprising, since Soil Taxonomy is based on full soil profile characteristics and taxonomic properties that purposefully omit dynamic surface layer characteristics to focus on the subsurface horizons indicative of longer-term pedogenic process (Forbes, 1986). The only dynamic surface layer characteristic used to distinguish a mineral soil order in Soil Taxonomy is OM content and color used to define a mollic epipedon, one of the main, but not the only, criteria necessary to be a Mollisol. This is not to say that Soil Taxonomy is necessarily deficient—it is just suited to a purpose other than soil health research, which is by definition focused on the more dynamic soil properties influenced by human management near the surface (Bunemann et al., 2018). As this research has shown, Soil Taxonomy often leads to a high-level grouping of soils distinguished more by diagnostics of pedogenic process than substantial differences in soil properties linked to soil function and

performance for agriculture (Figure 6e–f, and Supplemental Figure S2). Because of the hierarchical nature of Soil Taxonomy, it follows logically that if the highest level (soil order) is not optimal for diagnosing soil health, then intermediate levels of Soil Taxonomy (e.g., suborder, great group) will also fall short, leading to an overly complicated classification effort with too many classes.

There have been continued calls from pedologists to better document and understand humanity as a sixth soil-forming factor to detail how human activities alter soils (Richter, 2007, 2020; Yaalon & Yaron, 1966) and their relationship to soil health (quality) (Norfleet et al., 2003; Tugel et al., 2005). Similarly, the proposed concepts of genoform and phenoform demonstrate that pedologists recognize that soils of the same development pathways (genoforms) can take on different forms and properties as a result of human management (phenoforms) (Rossiter & Bouma, 2018). However, this research highlights that a number of soils from different USDA-NRCS soil taxonomic orders (genoforms) have similar 0-to-30-cm properties and depths to restrictive horizons, irrespective of soil development pathways. Unraveling anthropogenic forcing of semidynamic soil properties within diverse landscapes could be more fruitful if the classification framework used to guide soil health studies is based on a broad suite of soil properties tied to soil function, such as those used in this research.

4.3 | Soil performance limitations: Linking soil health regions to optimal soil health practices

This research has proposed a regional framework for interpreting and guiding soil health management. Research is beginning to show that the “success” of soil health practices is place based. A recent global meta-analysis of cover crop effects on SOC shows that texture mediates cover cropping’s impact: fine-textured soils showed a mean 39.5% increase in SOC compared with smaller increases in cover-cropped, coarse-textured, and medium-textured soils of 10–11% (Jian et al., 2020). In a long-term study of a cotton (*Gossypium hirsutum* L.)–tomato (*Solanum lycopersicum* L.) rotation in the relatively dry western San Joaquin Valley (in Region 2, “loamy with no restrictions”), cover cropping (with some supplemental irrigation) and reduced tillage (40–50% decrease in tractor operations) increased 0-to-30-cm SOC stocks by 5 Mg ha^{−1} after 8 yr and 10 Mg ha^{−1} after 14 yr relative to a control (Mitchell et al., 2015, 2017). Pretrial SOC stocks increased by 9 Mg ha^{−1} in this soil (23–32% clay, 23–33% silt, and 35–54% sand) even in the control (conventional tillage and no cover crops). Would the same results have occurred in an arid location in Region 1 (“coarse with no restrictions”) and with no supplemental irrigation? Generally, how would

available soil moisture to winter cover crops constrain the benefits of annual cover crop production across the north-to-south precipitation gradient in this study area (Figure 4, Table 3)? Would even greater SOC benefits accrue by cover cropping the shrink–swell region, where 30% of California’s annual field crops, 15% of small fruit and vegetables, and 10% of orchards are cultivated (Table 4)? Although only 4% of California’s vineyards are grown in this region (Table 4), 230,000 ha of orchards in the shrink–swell region may also especially benefit from cover crop induced drainage and aeration improvements of a naturally slow-to-drain landscape. For example, the soils at the western San Joaquin Valley research site mentioned above showed twofold faster infiltration rates when cover cropped (Mitchell et al., 2017). It might also be possible that Region 6 (“fine salt-affected”) is undersaturated with respect to SOC (Figure 5b) compared with regions with similar soil textures (Figure 5a), on account of its lower natural productivity (historically lower net primary production feeding SOC stocks). Cultivation of salt-tolerant cover crops could supply additional biomass for building soil carbon reserves.

Similar to cover crops, compost application guidelines should be tailored to soil health regions, especially if a primary goal is medium- to long-term sequestration of carbon. On-farm soil health research in California has shown that farmers are biased towards applying organic amendments on “poorer quality” soils (Andrews et al., 2003). However, this logical and intuitive approach to rationing limited resources may not be an optimal strategy. Soil organic carbon stabilization is greater in soils with more specific surface area (Six, Conant, Paul, & Paustian, 2002; von Lutzow et al., 2006), a soil property directly tied to clay content. Moreover, since soil aggregates are more stable in finer-textured soils, sequestration of carbon from applied compost could be greater in the loamier and finer textured soil health regions, covering about half of the study area (Table 2). In a California soil health research trial in double-cropped vegetable production on a relatively coarse soil (70% sand, 22% silt, and 8% clay), 8 yr of compost applications (36 Mg total C ha^{−1}) and various annual cover crops (24–35 Mg total C ha^{−1}) did not compensate for a dramatic first-year decline in SOC (31 Mg ha^{−1}) attributed to intensive tillage (White, Brennan, Cavigelli, & Smith, 2020). In spite of the study site’s relatively cool coastal climate, the ameliorative effect of compost and annual cover cropping on total SOC was only +9 and +3 Mg C ha^{−1}, respectively, but perhaps a completely different result would have been observed on a finer textured soil?

Interpretation of a farm’s soil OM levels from the context of a soil health region could allow farmers and policy makers to better understand where farms lie with respect to expectations, especially whether or not a location has reached carbon saturation of their soil (Stewart, Paustian, Conant,

Plante, & Six, 2007). However, since each of these regions cover broad swaths from northern to southern ends of the study area, research on temperature effects on SOC sequestration from applied compost could also help optimize use of this resource. Specifically, it is hypothesized that compost applications for SOC sequestration would make the most sense in northern shrink–swell regions and the least sense in coarser soils of the warmer, southern half of the Central Valley (Figure 4, Table 3). However, research has not yet examined how the sequestration levels may vary across different soils except in short-term controlled studies that have shown higher initial respiration of applied compost in soils with lower clay compared with higher clay content (Duong, Penfold, & Marschner, 2012). In the salt-affected regions, high-rate ($>100 \text{ Mg ha}^{-1}$) compost applications, such as those recently tested in Bay Area rangelands (Ryals & Silver, 2013), need to be carefully evaluated, because large compost applications would also include a substantial dose of salt. However, agronomic rates of $10\text{--}20 \text{ Mg compost ha}^{-1}$ demonstrated benefits of compost to outweigh deleterious impacts on soil salinity across a range of textures and crops in greenhouse media experiments (Reddy & Crohn, 2012).

5 | CONCLUSION

This research has revealed an inherent structure in SSURGO data as a means to guide soil health investigations and practices. Data from near-surface soil properties and root restrictive horizons were used to derive a seven-region soil health identity classification in an agricultural region of global importance. The essential concept is a landscape evenly divided between major agronomic performance limitations (alkalinity and salt-affected chemistry, shrink–swell characteristics, and/or root restrictive horizons) and soils with relatively homogenous profiles and no major physical or chemical limitations for agriculture (Figures 3–6). These regions are proposed to help inform soil health diagnostic interpretations and provide a place-based platform for the evaluation of management practices believed to promote soil health. Given the lack of publicly available data, the map is a starting point, a relative improvement over Soil Taxonomy, and is presented as a hypothesis to guide and help interpret future soil health research and diagnostics. This research has laid the framework for a general understanding and appreciation for the diversity of soil systems in California, each with characteristic soil properties and environmental conditions. We believe these regions can help guide soil health research and develop a fine-tuned toolbox of place-based approaches suitable to the limitations defined by nature, to the multiplicity of functions provided by soil's underlying vitality, and to the ever-evolving diversity of human goals.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Scott M. Devine: Formal analysis; Methodology; Validation; Visualization; Writing-original draft. Kerri L. Steenwerth: Conceptualization; Funding acquisition; Project administration; Supervision; Writing-review & editing. Anthony T. O'Geen: Conceptualization; Funding acquisition; Methodology; Project administration; Supervision; Visualization; Writing-review & editing.

DATA AVAILABILITY STATEMENT

Publicly available geospatial and tabular data used in this research are also available from the corresponding author upon request, including the soil health region delineations. The R code used in this research is available at <https://github.com/smdevine/SoilHealthRegions>.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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